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by Marschall, Jürgen

in: Manuscripta mathematica, (page(s) 47 - 66)
Berlin, Heidelberg; 1968

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manuscripta math. 58, 47 - 65 (1987)

manuscripta mathematica
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THE TRACE OF SOBOLEV-SLOBODECKIJ SPACES ON LIPSCHITZ DOMAINS

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In this paper we give some well-known trace theorems for Sobolev-Slobodeckij spaces under minimal regularity assumptions on the domain.

1. Introduction and Main Results

Let $\Omega \subset \mathbb{R}^n$ be a domain. For $s = 0,1,2,\ldots$ and $1 \le p \le \infty$ consider the Sobolev space $W^{sp}(\Omega)$ of all $f \in L^p(\Omega)$ such that the distributional derivatives $\partial^\alpha f$ of order $|\alpha| \le s$ belong to $L^p(\Omega)$. The norm on $W^{sp}(\Omega)$ is defined by

$$\|f\|_{W^{sp}(\Omega)} := \sum_{|\alpha| \le s} \|\partial^\alpha f\|_{L^p(\Omega)} . \tag{1}$$

When s is fractional, i.e. when $s = k + \lambda$ with $k \in \mathbb{N}$ and $0 < \lambda < 1$ let the Slobodeckij space $W^{sp}(\Omega)$ be the space of all $f \in W^{kp}(\Omega)$ such that

$$\|f\|_{W^{sp}(\Omega)} := \|f\|_{W^{kp}(\Omega)} + \sum_{|\alpha|=k} \Big(\int_\Omega \int_\Omega \frac{|\partial^\alpha f(y) - \partial^\alpha f(x)|^p}{|y-x|^{n+\lambda \cdot p}} \, dy dx \Big)^{\frac{1}{p}} < \infty . \tag{2}$$

When $p = \infty$, W^{sp} is, of course, the Hölder space, i.e. $W^{s\infty}(\Omega) = C^{k,\lambda}(\Omega)$.

The open set Ω is called a special Lipschitz domain, if there is a coordinate system and a Lipschitz function $\varphi : \mathbb{R}^{n-1} \to \mathbb{R}$ such that in this coordinate system

$$\Omega = \{(x',x_n) : x_n > \varphi(x')\} .$$

The boundary $\partial\Omega$ is then $\{(x',\varphi(x')) : x' \in \mathbb{R}^{n-1}\}$.

Ω is called a Lipschitz domain or minimally smooth, if there exist an $\varepsilon > 0$, a natural number N, a constant $M > 0$ and a sequence U_j of open sets such that:

(i) If $x \in \partial\Omega$, then $B(x,\varepsilon) \subset U_j$ for some j.

(ii) At most N of the U_j's have nonempty intersection.

(iii) For j there exists a special Lipschitz domain Ω_j such that

(iii_1) $U_j \cap \Omega = U_j \cap \Omega_j$

(iii_2) $\| \nabla\varphi_j \|_{L^\infty} \leq M$

when Ω_j is defined by φ_j. If instead of (iii_2) we have

(iii_2') $\| \nabla\varphi_j \|_{C^{k-1,\lambda}} \leq M$,

then we say that Ω is a $C^{k,\lambda}$-domain.

Minimal smooth domains are introduced in Stein's book [8]. One basic property is that $D_\Omega(\mathbb{R}^n)$, i.e. the restriction of test functions to Ω, is dense in $W^{sp}(\Omega)$ for $1 \leq p < \infty$. Another important feature is that there exists a bounded linear extension operator

$$E : W^{sp}(\Omega) \to W^{sp}(\mathbb{R}^n) \tag{3}$$

which is independent of $s \geq 0$ and $1 \leq p \leq \infty$. For $s = 0,1,2...$ this operator is constructed in [8], Chapter 6. The general case then follows by real interpolation (see [9], Chapter 4).

In the following we assume that Ω is a Lipschitz domain. Any $f \in D_\Omega(\mathbb{R}^n)$ has a well defined trace $\mathrm{Tr}\, f := f\big|_{\partial\Omega}$. This operator extends by continuity to

$$\mathrm{Tr} : W^{sp}(\Omega) \to L^p(\partial\Omega) , \tag{4}$$

if $s > \frac{1}{p}$ (see [6], [7] and [9]). Here $\partial\Omega$ carries the surface measure $dS = (1+|\nabla\varphi|^2)^{\frac{1}{2}}\, dx'$. For a more sophisticated definition of the trace see Section 2 below.

Let $n := (1+|\nabla\varphi|^2)^{-\frac{1}{2}} (\nabla\varphi, -1)$ be the outer unit normal to $\partial\Omega$ and consider the normal derivatives

$$\frac{\partial^k}{\partial n^k} u := \sum_{|\alpha|=k} \frac{k!}{\alpha!} \partial^\alpha u\, n^\alpha \tag{5}$$

for $u \in D_\Omega(\mathbb{R}^n)$. Then by (4) the mapping

$$\mathrm{Tr}_k\, u := \{ u\big|_{\partial\Omega}, \frac{\partial u}{\partial n}, \ldots, \frac{\partial^{k-1} u}{\partial n^{k-1}} \}$$

extends to a bounded linear operator

$$\mathrm{Tr}_k : W^{sp}(\Omega) \to \bigtimes_{i=0}^{k-1} L^p(\partial\Omega) , \tag{6}$$

if $s > k - \frac{1}{p'}$.

For applications to partial differential equations it is important to know the kernel and the image of the trace operator Tr_k. If Ω is sufficiently smooth, both are known for a long time (see [6], [7] and [9]). Denote by $W_0^{sp}(\Omega)$ the closure of the test functions $C_0^\infty(\Omega)$ in $W^{sp}(\Omega)$.

Theorem 1

Let Ω be a bounded or unbounded Lipschitz domain. Suppose that $1 < p < \infty$, $s > \frac{1}{p}$ and $k - \frac{1}{p'} < s \le k + \frac{1}{p}$, $\frac{1}{p} + \frac{1}{p'} = 1$. Then we have
$\mathrm{kernel}\,(\mathrm{Tr}_k) = W_0^{sp}(\Omega)$.

□

Note that the theorem answers a question posed by Nečas (see [7]). In case s an integer there is a closely related result by Hedberg and Wolff (see [2]).

For the determination of the image of the trace operator let us first define Besov spaces on the boundary. Let Ω be a Lipschitz domain, and in case $s \geq 1$ suppose that Ω is a $C^{k,\lambda}$-domain, $s > k + \lambda$. Suppose first that Ω is a special Lipschitz domain. Then when s is not an integer define $B_p^s(\partial\Omega)$ by transporting $W^{sp}(\mathbb{R}^{n-1})$ to $\partial\Omega$ via the mapping $(x',0) \to (x',\varphi(x'))$. In case s an integer transport the Besov space $B_p^s(\mathbb{R}^{n-1})$ defined by

$$\|f\|_{B_p^s} := \|f\|_{W^{s-1,p}} + \sum_{|\alpha|=s-1} \Big(\int\int \frac{|\partial^\alpha f(x+h)-2\partial^\alpha f(x)+\partial^\alpha f(x-h)|^p}{|h|^{n+p}}\, dxdh \Big)^{\frac{1}{p}} < \infty . \tag{7}$$

In case of a general Lipschitz domain reduce the definition of $B_p^s(\partial\Omega)$ with a partition of unity to the case of a special Lipschitz domain (compare [8], Chapter 6). Then there is the

Theorem 2

Let Ω be a bounded or unbounded Lipschitz domain. Suppose that $1 < p < \infty$ and $s > \ell - \frac{1}{p'}$. In case $s - \frac{1}{p} \geq 1$ suppose in addition that Ω is a $C^{k,\lambda}$-domain, $k + \lambda > s - \frac{1}{p}$. Then the mapping

$$Tr_\ell : W^{sp}(\Omega) \to \prod_{j=0}^{\ell-1} B_p^{s-\frac{1}{p}-j}(\partial\Omega)$$

is a surjection. In case $s - \frac{1}{p}$ not an integer it has a bounded linear right inverse.

□

Both theorems are well-known when $k + \lambda > s + 1$. The essential point here is that they already hold under weakest assumptions. In case of Theorem 1 this means that Ω has a Lipschitz boundary, and in case of Theorem 2 this means that the boundary spaces $B_p^s(\partial\Omega)$ are well defined.

We don't know, if Tr_ℓ has a bounded linear right inverse in case $s - \frac{1}{p}$ an integer. For first steps in this direction see [1] and [3].

We remark that the theorems also hold for $p = 1$, if we exclude the case $s \in \mathbb{N}$. In fact, both theorems are true for more general Besov and potential spaces. Our method applies to these spaces with minor modifications. We have decided to deal only with Sobolev-Slobodeckij spaces merely to make matters simple.

The question arises what is the description of the trace space in case of a Lipschitz boundary. The answer is, of course, given in the papers by Alf Jonsson and Hans Wallin (see e.g. [4]). They described the trace space for d-sets. However, in case of a Lipschitz boundary their descriptions simplify greatly.

Suppose that s is not an integer. Define $\tilde{V}^{sp}(\partial\Omega)$ to be the space of all $\{f_\alpha\} \subset \underset{|\alpha| < s}{X} L^p(\partial\Omega)$ such that

$$\|\{f_\alpha\}\|_{\tilde{V}^{sp}(\partial\Omega)} := \sum_{|\alpha| < s} \|f_\alpha\|_{L^p(\partial\Omega)} + \sum_{|\alpha| = [s]} \left(\int\int \frac{|f_\alpha(y) - f_\alpha(x)|^p}{|y-x|^{n+(s-[s])p}} \, dydx \right)^{\frac{1}{p}} < \infty . \tag{8}$$

There is a natural mapping

$$f \to \{\partial^\alpha f\} : D_\Omega(\mathbb{R}^n) \to \tilde{V}^{sp}(\partial\Omega) .$$

Let $V^{sp}(\partial\Omega)$ be the closure of the image of $D_\Omega(\mathbb{R}^n)$ in $\tilde{V}^{sp}(\partial\Omega)$.

Theorem 3

Let $1 < p < \infty$ and $k - \frac{1}{p} < s < k + \frac{1}{p}$, $s > \frac{1}{p}$. Then the mapping

$$f \to \{\partial^\alpha f\} : W^{sp}(\Omega) \to V^{s - \frac{1}{p}, p}(\partial\Omega)$$

is surjective and has a bounded linear right inverse.

□

Observe that by the definition of $\tilde{V}^{sp}(\partial\Omega)$ the f_α are independent of each other. Hence, it is surprising that $V^{s-\frac{1}{p},p}(\partial\Omega)$ is the trace space of $W^{sp}(\Omega)$. The Lipschitz condition is strong enough to guarantee this.

Let us give the plan of the rest of the paper. In Section 2 we describe the results obtained by Alf Jonsson and Hans Wallin. They are essential for the proofs of our theorems. In Section 3 we prove a lemma which gives the link between the results by Jonsson/Wallin and our problems. And in the final section we prove the theorems.

2. The Trace Spaces in Lipschitz Domains

Jonsson and Wallin studied Besov spaces on general d-sets. The important point is that the boundary $\partial\Omega$ of a Lipschitz domain with surface measure is a (n-1)-set (see [5], Section 2.1). In the following we describe the results obtained by Jonsson and Wallin applied to the boundary of a Lipschitz domain.

The definition of the spaces $\tilde{B}^s_p(\partial\Omega)$ is analogous to the definition of Lipschitz spaces by Whitney (compare [8], Chapter 6). Let $s = k + \lambda$, $k \in \mathbb{N}$ and $0 < \lambda < 1$. Let $\{f_\alpha\}_{|\alpha| \le k} \in X\, L^p(\partial\Omega)$ and define

$$f_\alpha(x) = \sum_{|\alpha+\beta| \le k} \frac{1}{\beta!}(x-y)^\beta f_{\alpha+\beta}(y) + R_\alpha(x,y) \tag{9}$$

for $x,y \in \partial\Omega$. Then we have $\{f_\alpha\} \in \tilde{B}^s_p(\partial\Omega)$ iff

$$\begin{aligned} \|\{f_\alpha\}\|_{\tilde{B}^s_p(\partial\Omega)} := \sum_{|\alpha| \le k} \Big(& \|f_\alpha\|_{L^p(\partial\Omega)} \\ & + \Big(\int\int_{|x-y|<1} \frac{|R_\alpha(x,y)|^p}{|x-y|^{n-1+(s-|\alpha|)p}}\, dS_y dS_x \Big)^{\frac{1}{p}} \Big) < \infty . \end{aligned} \tag{10}$$

When $s = k + 1$, $\tilde{B}^s_p(\partial\Omega)$ is defined by approximation of smoother functions

Let $s < s^* < k + 2$. We say $\{f_\alpha\}_{|\alpha| \le k} \in \tilde{B}_p^{k+1}(\partial\Omega)$, if there exist $\{g_{\alpha j}\}_{|\alpha| \le k+1} \in \tilde{B}_p^{s^*}(\partial\Omega)$, $j = 1,2,3\ldots$, such that for some sequence $\{a_j\} \in \ell^p$ the following conditions are satisfied:

$$\left.\begin{aligned} &\| f_\alpha - g_{\alpha j} \|_{L^p(\partial\Omega)} \le 2^{-j(k+1-|\alpha|)} a_j, \quad |\alpha| \le k , \\ &\| g_{\alpha j} - g_{\alpha, j+1} \|_{L^p(\partial\Omega)} \le a_j, \quad |\alpha| = k + 1 , \\ &\| \{g_{\alpha j}\} \|_{\tilde{B}_p^{s^*}(\partial\Omega)} \le 2^{-j(k+1-s^*)} a_j . \end{aligned}\right\} \tag{11}$$

It turns out that $\tilde{B}_p^{s - \frac{1}{p}}(\partial\Omega)$ is the trace space of $W^{sp}(\Omega)$. First let us give a precise meaning to the concept of trace.

A locally integrable function f is said to be strictly defined, if

$$\overline{f}(x) = \lim_{r \to 0} \frac{1}{|B|} \int_{B(x,r)} f(y)dy , \tag{12}$$

at all points where this limit exist. By a famous theorem by Lebesgue we have $\overline{f} = f$ a.e. with respect to Lebesgue measure. But in case $f \in W^{sp}(\mathbb{R}^n)$ much more is true.

Lemma 4

Let $f \in W^{sp}(\mathbb{R}^n)$ be such that $s > k + \frac{1}{p}$. Then for $|\alpha| \le k$ $\partial^\alpha f$ can be strictly defined a.e. with respect to (n-1)-dimensional Hausdorff measure.

□

For the proof of the lemma and for the definition of Hausdorff measure see [5], Section 2.2. The important point is that the restriction of the (n-1)-dimensional Hausdorff measure to $\partial\Omega$ is equivalent to the surface measure (see [5], Proposition 1.1). Hence, under the hypothesis of the lemma $\overline{\partial^\alpha f}\big|_{\partial\Omega}$ exist a.e. with respect to the surface measure. And it turns out that

$$\mathrm{Tr}(\partial^\alpha f) = \overline{\partial^\alpha f}\Big|_{\partial\Omega} . \tag{13}$$

Theorem 5
Let $1 < p < \infty$ and $k - \frac{1}{p'} < s \leq k + \frac{1}{p}$, $s > \frac{1}{p}$.

(i) The mapping

$$f \to \{Tr(\partial^\alpha f)\}_{|\alpha| \leq k-1} : W^{sp}(\mathbb{R}^n) \to \tilde{B}_p^{s - \frac{1}{p}}(\partial\Omega)$$

is a bounded surjection.

(ii) In case $k - \frac{1}{p'} < s < k + \frac{1}{p}$ it has a bounded linear right inverse

$$E_k : \tilde{B}_p^{s - \frac{1}{p}}(\partial\Omega) \to W^{sp}(\mathbb{R}^n) .$$

(iii) In case $s - \frac{1}{p}$ an integer there exists a bounded nonlinear right inverse $\tilde{E}_k$.

(iv) One has $E_k(\{f_\alpha\}), \tilde{E}_k(\{f_\alpha\}) \in C^\infty(\mathbb{R}^n \setminus \partial\Omega)$.

□

The proof is found in [4], Theorems 6.1, 6.2, 6.3, 7.1, 7.2 and 7.3. (13) is part of the proof.

We will need a further property of the extension operator.

Lemma 6
Let $f \in C_0^\infty(\mathbb{R}^n)$. We then have

$$E_k(\{\partial^\alpha f\}_{|\alpha| \leq k-1}), \tilde{E}_k(\{\partial^\alpha f\}_{|\alpha| \leq k-1}) \in C_0^{k-1,1}(\mathbb{R}^n) .$$

Proof
The statement for $\tilde{E}_k$ follows from the case $p = \infty$ and $q = 1$ of Theorem 6.3 in [4]. The assertion for E_k is not stated explicitly there. However, E_k is an extension operator of Whitney type. Thus, one may copy the proof of Theorem 6.4 in [8].

□

3. A Fundamental Lemma

By (3) we may suppose that any $f \in W^{sp}(\Omega)$ is defined on $\mathbb{R}^n$, i.e. $f \in W^{sp}(\mathbb{R}^n)$. Then by (13) and Theorem 5 we have

$$\frac{\partial^\ell}{\partial n^\ell} f = \sum_{|\alpha|=\ell} \frac{\ell!}{\alpha!} \overline{\partial^\alpha f}\, n^\alpha \tag{14}$$

for $\ell < s - \frac{1}{p}$. Conversely, if $\{f_\alpha\} \in \tilde{B}_p^{k+\lambda}(\partial\Omega)$, $0 < \lambda \leq 1$, we may define

$$\frac{\partial^\ell}{\partial n^\ell}\{f_\alpha\} := \sum_{|\alpha|=\ell} \frac{\ell!}{\alpha!} f_\alpha\, n^\alpha \tag{15}$$

and $Tr_{k+1}\{f_\alpha\} := \{f_o, \frac{\partial}{\partial n}\{f_\alpha\}, \ldots, \frac{\partial^k}{\partial n^k}\{f_\alpha\}\}$. Then (14) shows that this is consistent with the definition in Section 1. Now the main result in this section is the

Lemma 7
The mapping

$$Tr_{k+1} : \tilde{B}_p^{k+\lambda}(\partial\Omega) \to \bigtimes_{\ell=o}^{k} L^p(\partial\Omega)$$

is injective.

□

When Ω is a sufficiently smooth domain the lemma is proved by choosing normal coordinates and thus transfering the problem to a half space. However, in case of a Lipschitz domain we have only one derivative at our disposal. Hence, a different proof is required. We need the

Lemma 8
Let $s > 1 + \frac{1}{p}$ and Ω be a special Lipschitz domain. Then the mapping

$$f \to \overline{f}(x', \varphi(x')) : W^{sp}(\Omega) \to W^{1p}(\mathbb{R}^{n-1})$$

is bounded. Moreover, for $j = 1, \ldots, n-1$ we have

$$\frac{\partial}{\partial x_j} \overline{f}(x', \varphi(x')) = \overline{\frac{\partial f}{\partial x_j}}(x', \varphi(x')) + \overline{\frac{\partial f}{\partial x_n}}(x', \varphi(x'))\, \frac{\partial \varphi}{\partial x_j}(x') \tag{16}$$

in the weak sense.

Proof

The fact that the mapping $f \to \overline{f}(x',\varphi(x'))$ is well defined and bounded, is just a standard trace theorem (compare e.g. [9], Chapter 4). Also (15) holds, if $f \in W^{sp}(\mathbb{R}^n) \cap C^1(\mathbb{R}^n)$ (recall that we may consider f as defined on all of $\mathbb{R}^n$).

Hence, approximate an arbitrary f by

$$f_r(x) := \frac{1}{|B|} \int_{B(x,r)} f(y)dy \ .$$

Then we have $f_r \in W^{sp} \cap C^1$ and $f_r \to f$ in the norm of W^{sp}, hence, $\frac{\partial}{\partial x_j} f_r(x',\varphi(x')) \to \frac{\partial}{\partial x_j} \overline{f}(x',\varphi(x'))$ in the norm of $L^p(\mathbb{R}^{n-1})$. Now observe that

$$\frac{\partial}{\partial x_i} f_r = \left(\frac{\partial f}{\partial x_i} \right)_r = \frac{1}{|B|} \int_{B(x,r)} \frac{\partial f}{\partial x_i} dy$$

for each $x \in \mathbb{R}^n$. Hence, (16) follows from Lemma 4.

□

We actually need the following consequence of the lemma.

Corollary 9

If $\overline{f}(x',\varphi(x')) \equiv 0$, then for $j = 1,\ldots,n-1$

$$\overline{\frac{\partial f}{\partial x_j}}(x',\varphi(x')) + \overline{\frac{\partial f}{\partial x_n}}(x',\varphi(x')) \frac{\partial \varphi}{\partial x_j}(x') \equiv 0 \ .$$

□

Proof of Lemma 7

The lemma is plain, if $k = 0$. Hence, proceed by induction on k. Thus, suppose that $Tr_{k+1}\{f_\alpha\} = 0$. By induction hypothesis we have $f_\alpha \equiv 0$ for $|\alpha| \le k - 1$. By Theorem 5 there is an $f \in W^{k+\lambda+\frac{1}{p},p}(\mathbb{R}^n)$ such that $\overline{\partial^\alpha f}\Big|_{\partial\Omega} = f_\alpha$ for all $|\alpha| \le k$. Hence, apply the corollary to each f_α such that $|\alpha| = k - 1$. The trick is to write the resulting equations as $A(\{f_\alpha\}) = 0$ with a nonsingular matrix A. This obviously yields the lemma.

Now order $\{f_\alpha\}_{|\alpha|=k}$ in any way by increasing α_n such that the last components are $f_{(1,\ldots,k-1)},\ldots,f_{(o,\ldots,1,k-1)}, f_{(o,\ldots,k)}$. For abbreviation let $a_j := \frac{\partial\varphi}{\partial x_j}$. Then the resulting matrix has the following form:

$$
\begin{array}{cccccccccccccccc}
1 & * & . & . & . & . & . & . & . & . & . & . & . & * & 0 \\
0 & . & . & & & & & & & & & & & . & . \\
. & . & . & . & & & & & & & & & & . & . \\
. & & . & . & . & & & & & & & & & . & . \\
. & & & . & . & . & & & & & & & & . & . \\
. & & & & . & . & . & & & & & & & . & . \\
. & & & & & . & . & . & & & & & & . & . \\
. & & & & & & . & . & * & . & . & . & . & * & 0 \\
. & & & & & & & . & 1 & 0 & . & . & . & 0 & a_1 \\
. & & & & & & & & . & . & . & & & . & . \\
. & & & & & & & & & . & . & . & & . & . \\
. & & & & & & & & & & . & . & . & . & . \\
. & & & & & & & & & & & & . & 0 & . \\
0 & . & . & . & . & . & . & . & . & . & . & . & . & 0 & 1 & a_{n-1} \\
* & . & . & . & . & . & . & * & \gamma_1 & . & . & . & . & \gamma_{n-1} & (-1)^k &
\end{array}
$$

The last row where we have defined $\gamma_j := (-1)^{k-1}\, k a_j$ comes from the equation $\frac{\partial^k}{\partial n^k}\{f_\alpha\} = 0$.

The point is now the following: If α is such that $\alpha_j \neq 0$ and $\alpha_n < k$, then by applying the Gauß algorithm we add to the last row terms of the form

$$
\begin{aligned}
& -\frac{k!}{\alpha!}\, n^\alpha\, f_\alpha - \frac{k!}{\alpha!}\, n^\alpha\, a_j\, f_\beta \\
&= -\frac{k!}{\alpha!}\, n^\alpha\, f_\alpha + \Big(\frac{\max\{1,\alpha_n\}}{\alpha_j}(1+|\nabla\varphi|^2)^{-\frac{1}{2}}\, a_j^2\Big)\, \frac{k!}{\beta!}\, n^\beta\, f_\beta
\end{aligned}
$$

with $\beta := \{\alpha_1,\ldots,\alpha_j-1,\ldots,\alpha_n+1\}$. But this means that in each step we multiply $\frac{k!}{\beta!}\, n^\beta\, f_\beta$ by a factor $\lambda \geq 1$. Hence, A cannot be singular, and the lemma is proved.

□

4. Proofs of the Theorems

The proofs of the theorems are now easy.

Proof of Theorem 1

(i) The inclusion $W_0^{sp}(\Omega) \subset \text{kernel } (Tr_k)$ is obvious. For the other direction observe that, if $f \in \text{kernel } (Tr_k)$, then by Lemma 7

$$\overline{\partial^\alpha f}\Big|_{\partial\Omega} \equiv 0 \text{ for all } |\alpha| \le k - 1.$$

Choose a sequence $f_j \in C_0^\infty(\mathbb{R}^n)$ such that $f_j \to f$ in $W^{sp}(\mathbb{R}^n)$. Then by Theorem 5 (i) $\{\partial^\alpha f_j\} \to 0$ in $\tilde{B}_p^{s-\frac{1}{p}}(\partial\Omega)$. Define $g_j := E_k(\{\partial^\alpha f_j\})$ (resp. $g_j := \tilde{E}_k(\{\partial^\alpha f_j\})$. Since the extension operator is bounded, we have $g_j \to 0$ in $W^{sp}(\mathbb{R}^n)$. Moreover, by Lemma 6 $g_j \in C_0^{k-1,1}(\mathbb{R}^n)$. Hence, defining $h_j := f_j - g_j$ the the proof is reduced to the case $f \in \text{kernel } (Tr_k) \cap C_0^{k-1,1}$.

(ii) There is no loss of generality to assume that Ω is bounded. For $f \in \text{kernel } (Tr_k) \cap C^{k-1,1}$ the Taylor formula implies

$$|\partial^\alpha f(x)| \le Cd(x,\partial\Omega)^{k-|\alpha|}, \quad |\alpha| \le k . \tag{17}$$

Given $\varepsilon > 0$ there is $\varphi_\varepsilon \in C_0^\infty(\Omega)$ such that $\varphi_\varepsilon(x) = 1$ for $x \in \Omega$ with $d(x,\partial\Omega) \ge \varepsilon$ and such that for any α

$$|\partial^\alpha \varphi_\varepsilon(x)| \le C_\alpha \varepsilon^{-|\alpha|} . \tag{18}$$

Define $f_\varepsilon := \varphi_\varepsilon \cdot f$. Then obviously $f_\varepsilon \to f$ in W^{kp}. Hence, suppose that $s = k + \lambda$, $0 < \lambda \le \frac{1}{p}$. We claim that for any α with $|\alpha| = k$

$$\int_\Omega \int_\Omega \frac{|\partial^\alpha(\psi_\varepsilon f)(x) - \partial^\alpha(\psi_\varepsilon f)(y)|^p}{|x-y|^{n+\lambda p}} \, dxdy \le C \max\{\delta_\varepsilon, \varepsilon^{1-\lambda p}\} . \tag{19}$$

Here we have defined $\psi_\varepsilon := 1 - \varphi_\varepsilon$ and δ_ε is such that $\delta_\varepsilon \to 0$ for $\varepsilon \to 0$.

(iii) Let $\Omega_\delta := \{x \in \Omega : d(x,\partial\Omega) \le \delta\}$. Then the integration in (19) is performed over $\Omega_{2\varepsilon} \times \Omega_{2\varepsilon} \cup \Omega_{2\varepsilon} \times C\Omega_{2\varepsilon} \cup C\Omega_{2\varepsilon} \times \Omega_{2\varepsilon}$. In the complement of this union the integrand vanishes.

By (17) and (18) we have

$$\int_{\Omega_{2\varepsilon}} \int_{C\Omega_{2\varepsilon}} \frac{|\partial^\alpha(\psi_\varepsilon f)(x)|^p}{|x-y|^{n+\lambda p}}\,dxdy$$

$$\le C \int_{\Omega_\varepsilon} \int_{C\Omega_{2\varepsilon}} \frac{dxdy}{|x-y|^{n+\lambda p}}$$

$$\le C\,\varepsilon^{1-\lambda p}\ .$$

Hence, it remains to prove

$$\int_{\Omega_{2\varepsilon}} \int_{\Omega_{2\varepsilon}} \frac{|\partial^\alpha(\psi_\varepsilon f)(x) - \partial^\alpha(\psi_\varepsilon f)(y)|^p}{|x-y|^{n+\lambda p}}\,dxdy \le C \max\{\delta_\varepsilon, \varepsilon^{1-\lambda p}\}\ . \quad (20)$$

(iv) Let $\alpha = \alpha_1 + \alpha_2$ and consider

$$|\partial^{\alpha_1}\psi_\varepsilon(x)\partial^{\alpha_2}f(x) - \partial^{\alpha_1}\psi_\varepsilon(y)\partial^{\alpha_2}f(y)|$$

$$\le |\partial^{\alpha_1}\psi_\varepsilon(x)| \cdot |\partial^{\alpha_2}f(x) - \partial^{\alpha_2}f(y)|$$

$$+ |\partial^{\alpha_1}\psi_\varepsilon(x) - \partial^{\alpha_1}\psi_\varepsilon(y)| \cdot |\partial^{\alpha_2}f(y)|$$

$$= I + II\ .$$

We have $\|\psi_\varepsilon\|_{W^{\ell p}} \le C\,\varepsilon^{\frac{1}{p}-\ell}$. For $\ell = 0,1,2\ldots$ this follows from (18) and for arbitrary $\ell \ge 0$ then by interpolation. Hence, by (17) (20) holds for the term II.

If $\alpha_1 = 0$, then

$$\int_{\Omega_{2\varepsilon}} \int_{\Omega_{2\varepsilon}} |\psi_\varepsilon(x)|^p \frac{|\partial^\alpha f(x) - \partial^\alpha f(y)|^p}{|x-y|^{n+\lambda p}}\,dxdy \to 0$$

for $\varepsilon \to 0$, since $f \in W^{sp}$. Hence, there exist $\delta_\varepsilon > 0$ having the required property.

If $\alpha_1 \neq 0$, split the integration into $|x-y| \geq \varepsilon$ and $|x-y| < \varepsilon$. Then by (17) and (18)

$$\int_{\substack{\Omega_{2\varepsilon} \\ |x-y| \geq \varepsilon}} \int_{\Omega_{2\varepsilon}} |\partial^{\alpha_1}\psi_\varepsilon(x)|^p \frac{|\partial^{\alpha_2}f(x)-\partial^{\alpha_2}f(y)|^p}{|x-y|^{n+\lambda p}}\,dxdy \leq C\,\varepsilon^{1-\lambda p}$$

and since $\partial^{\alpha_2}f$ satisfies a Lipschitz condition

$$\int_{\substack{\Omega_{2\varepsilon} \\ |x-y| < \varepsilon}} \int_{\Omega_{2\varepsilon}} |\partial^{\alpha_1}\psi_\varepsilon(x)|^p \frac{|\partial^{\alpha_2}f(x)-\partial^{\alpha_2}f(y)|^p}{|x-y|^{n+\lambda p}}\,dxdy$$

$$\leq C\,\varepsilon^{-p} \int_{\substack{\Omega_{2\varepsilon} \\ |x-y| < \varepsilon}} \int_{\Omega_{2\varepsilon}} |x-y|^{-n+(1-\lambda)p}\,dxdy$$

$$\leq C\,\varepsilon^{1-\lambda p}\,.$$

Thus, (20) is proved, and we obtain (19).

(v) In case $0 < \lambda < \frac{1}{p}$ the theorem follows from (19). If $\lambda = \frac{1}{p}$, then $\psi_\varepsilon f$ remains bounded in W^{sp}. Since W^{sp} is reflexive, there is a weakly convergent subsequence $\psi_{\varepsilon_j} f$. Obviously $\psi_{\varepsilon_j} f \rightharpoonup 0$. But then by Mazur's theorem a convex combination of the $\psi_{\varepsilon_j} f$ converges strongly to zero. Hence, the theorem follows in case $s = k + \frac{1}{p}$, too.

□

We next prove Theorem 3, since we use it in the proof of Theorem 2.

Proof of Theorem 3

We may suppose that Ω is a special Lipschitz domain. Let $s = k + \lambda$ and $f \in D_\Omega(\mathbb{R}^n)$. We use a version of the Taylor formula due to Whitney (see [10]). For $x,y \in \partial\Omega$ let $z(t) := (x'+t(y'-x'), \varphi(x'+t(y'-x')))$, $0 \leq t \leq 1$ and $s = s(t)$ the arc length of the curve $z(t)$. Hence, the length is $L = s(1)$. We then have for $|\alpha| < k$

$$R_\alpha(x,y) = \partial^\alpha f(x) - \sum_{|\alpha+\beta| \le k} \frac{1}{\beta!}(x-y)^\beta \partial^{\alpha+\beta} f(y)$$

$$= \sum_{|\alpha+\beta|=k} \sum_{\substack{|\gamma|=1 \\ \gamma \le \beta}} \int_0^L (\partial^{\alpha+\beta} f(z(s)) - \partial^{\alpha+\beta} f(x)) \frac{(y-z(s))^{\beta-\gamma}}{(\beta-\gamma)!} \, dx_\gamma(s)$$

and hence,

$$|R_\alpha(x,y)| \le C \, |x-y|^{k-|\alpha|} \sum_{|\alpha+\beta|=k} \int_0^1 |\partial^{\alpha+\beta} f(z(t)) - \partial^{\alpha+\beta} f(x)| dt \, .$$

Then by Minkowski's integral inequality

$$\left(\int_{\partial\Omega} \int_{\partial\Omega} \frac{|R_\alpha(x,y)|^p}{|x-y|^{n-1+(s-|\alpha|)p}} \, dS_x dS_y \right)^{\frac{1}{p}}$$

$$\le C \sum_{|\alpha+\beta|=k} \int_0^1 \left(\int_{\partial\Omega} \int_{\partial\Omega} \frac{|\partial^{\alpha+\beta} f(z(t)) - \partial^{\alpha+\beta} f(x)|^p}{|x-y|^{n-1+\lambda p}} \, dS_x dS_y \right)^{\frac{1}{p}} dt$$

$$\le C \sum_{|\alpha+\beta|=k} \int_0^1 \left(\int_{\mathbb{R}^{n-1}} \int_{\mathbb{R}^{n-1}} \frac{|\partial^{\alpha+\beta} f(z(t)) - \partial^{\alpha+\beta} f(x)|^p}{|x'-y'|^{n-1+\lambda p}} \, dx' dy' \right)^{\frac{1}{p}} dt \, .$$

But then the substitution $y^* := (1-t)x' + ty'$ yields

$$\left(\int_{\partial\Omega} \int_{\partial\Omega} \frac{|R_\alpha(x,y)|^p}{|x-y|^{n-1+(s-|\alpha|)p}} \, dS_x dS_y \right)^{\frac{1}{p}}$$

$$\le C \sum_{|\alpha+\beta|=k} \left(\int_{\mathbb{R}^{n-1}} \int_{\mathbb{R}^{n-1}} \frac{|\partial^{\alpha+\beta} f(y^*, \varphi(y^*)) - \partial^{\alpha+\beta} f(x', \varphi(x'))|^p}{|y^*-x'|^{n-1+\lambda p}} \, dx' dy^* \right)^{\frac{1}{p}}$$

$$\le C \, \| f \|_{V^{sp}(\partial\Omega)} \, .$$

This proves the theorem.

□

Theorem 3 has an interesting corollary which we use in the proof of Theorem 2.

Corollary 10

Let $\Omega := \{x : x_n > \varphi(x')\}$ be a special Lipschitz domain. In case $s \geq 1$ suppose, in addition, that Ω is a $C^{k,\lambda}$-domain, $k + \lambda > s$. For a test function f define $\tilde{f}(x) := f(x', x_n - \varphi(x'))$. Then

$$\{\partial^\alpha f\} \to \{\partial^\alpha \tilde{f}\} : \tilde{B}^s_p(\mathbb{R}^{n-1}) \to B^s_p(\partial\Omega)$$

is a bounded mapping.

Proof

Suppose that $s > 1$ is not an integer. For $f \in C^\infty_o$ we have $\tilde{f} \in C^{k,\lambda}_o$. Then the proof of Theorem 3 shows that $\tilde{f} \in \tilde{B}^s_p(\partial\Omega)$. Moreover, using the chain rule one easily shows

$$\| \{\partial^\alpha \tilde{f}\} \|_{V^{sp}(\partial\Omega)} \leq C \, \| \{\partial^\alpha f\} \|_{V^{sp}(\mathbb{R}^{n-1})}$$

which implies the corollary in case s not an integer. But this implies the corollary in case s an integer, too (compare (11)).

□

Proof of Theorem 2

(i) We may suppose that $\ell - \frac{1}{p'} < s \leq \ell + \frac{1}{p}$. Then the theorem follows from Theorem 5, if we can show that

$$Tr_\ell : \tilde{B}^{s - \frac{1}{p}}_p(\partial\Omega) \to \bigtimes_{j=o}^{\ell-1} B^{s - \frac{1}{p} - j}_p(\partial\Omega) \qquad (21)$$

is an isomorphism.

We already know that Tr_ℓ is injective. Hence, it remains to show that it is surjective.

(ii) We may suppose that Ω is a special Lipschitz domain. The main step consists in showing that $\{f, \frac{\partial f}{\partial n}, \ldots, \frac{\partial^{\ell-1} f}{\partial n^{\ell-1}}\}$ is uniquely determined by $\{f_{(o,\ldots,o)}, \ldots, f_{(o,\ldots,\ell-1)}\}$ and conversely. We proceed by induction on ℓ.

We show that

$$g_j := \frac{\partial^j f}{\partial n_j} = (-1)^j \, (1+|\nabla\varphi|^2)^{\frac{j}{2}} \, f_{(o,\ldots,j)} + r_j \qquad (22)$$

where r_j depends only on $\partial^\alpha g_m$, $0 \le m \le j-1$ and $|\alpha| \le j-m$. This obviously implies our assertion.

Defining $a_i := \frac{\partial\varphi}{\partial x_i}$, $i = 1,\ldots,n-1$ and $a_n = -1$ we have

$$\sum_{|\alpha|=j} \frac{j!}{\alpha!} n^\alpha f_\alpha = (1+|\nabla\varphi|^2)^{-\frac{j}{2}} \sum_{|\alpha|=j} \frac{j!}{\alpha!} a^\alpha f_\alpha \, .$$

Now using Lemma 8 we get in case $\alpha_i \neq 0$

$$f_\alpha = r - a_i \, f_{(\alpha_1,\ldots,\alpha_{i-1},\ldots,\alpha_{n+1})}$$

where r is of the form r_j and hence,

$$f_\alpha = r + (-1)^{\alpha_1+\ldots+\alpha_{n-1}} \, a_1^{\alpha_1} \cdot \ldots \cdot a_{n-1}^{\alpha_{n-1}} \, f_{(o,\ldots,j)} \, .$$

Hence, we get

$$\begin{aligned} g_j &= (-1)^j \, (1+|\nabla\varphi|^2)^{-\frac{j}{2}} \Big(\sum_{|\alpha|=j} \frac{j!}{\alpha!} a^{2\alpha} \Big) \, f_{(o,\ldots,j)} + r_j \\ &= (-1)^j \, (1+|\nabla\varphi|^2)^{\frac{j}{2}} \, f_{(o,\ldots,j)} + r_j \end{aligned}$$

which was to be shown.

(iii) The preceeding step shows that there are $g_j' \in B_p^{s-\frac{1}{p}-j}(\partial\Omega)$ such that the solution of $g_j' = f_{(o,\ldots,j)}$ is equivalent to $g_j = \frac{\partial^j f}{\partial n^j}$.

Now define $\tilde{g}_j(x',0) := g_j'(x',\varphi(x'))$. Then obviously $\tilde{g}_j \in B_p^{s-\frac{1}{p}-j}(\mathbb{R}^{n-1})$. Hence, by the well-known trace theorems for the half space (see [9]) there is $\tilde{f} \in W^{sp}(\mathbb{R}^n_+)$ such that

$\frac{\partial^j}{\partial x_n^j} \tilde{f}(x',0) = \tilde{g}_j$, $j = 0,\dots,\ell-1$. But then $\{\tilde{f}_\alpha\} \in \tilde{B}_p^{s-\frac{1}{p}}(\mathbb{R}^{n-1})$.
Now the transformation of the corollary leaves $\tilde{f}_{(0,\dots,j)}$ fixed. Hence, there is $\{f_\alpha\} \in \tilde{B}_p^{s-\frac{1}{p}}(\partial\Omega)$ such that $f_{(0,\dots,j)} = g_j^!$. But by step (ii) this implies $\frac{\partial^j f}{\partial n^j} = g_j$, and the theorem is proved.

□

Let us conclude with a remark concerning the case $s - \frac{1}{p}$ an integer. Consider the function $\tilde{f} \in W^{sp}(\mathbb{R}^n_+)$ constructed in step (iii) of the proof. If Ω is a special $C^{k,\lambda}$-domain such that $k + \lambda > s$, then obviously $f(x) := \tilde{f}(x', x_n - \varphi(x')) \in W^{sp}(\Omega)$. This yields the

Corollary 11
Let $s - \frac{1}{p}$ an integer and suppose that Ω is a $C^{k,\lambda}$-domain such that $k + \lambda > s > \ell - \frac{1}{p}$. Then the mapping

$$Tr_\ell : W^{sp}(\Omega) \to \bigtimes_{j=0}^{\ell-1} B_p^{s-\frac{1}{p}-j}(\partial\Omega)$$

has a bounded linear right inverse.

□

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(Received September 1, 1986)